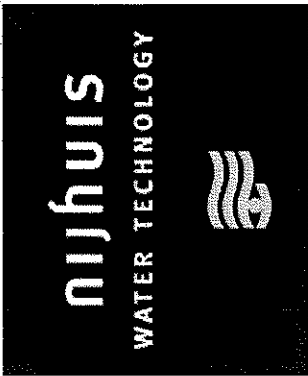


1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
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Client : Austral Ilo
Project name : Austral Ilo - 112048
Drawing number : 112048.145
Ordered by : Austral

Fabricated by (firm) : Nijhuis Water Technology B.V.
Dinxperlo, Holland
Path of project : H:\Projekten\...\Eplan\112048_A.p
Project name : Austral Ilo - 112048
Fabricate : Telemecanique, Allen Bradley, ABB, Omron/finder, Wieland
Type : Eldon or Rittal, coated steel panel
Power supply : 3x440V 60Hz
Location : Peru
Project manager : Sander Tielijes
Remarks : PLC controlled E-panel
Created at : 17.07.06
Last revision at : 13.03.13 by : PAV

Last page number : 78
Number of pages : 80

Project : Austral Ilo - 112048		Revision : A	Created on : 17.07.06	VISION ON SUSTAINABILITY Copyright reserved www.nijhuis-water.com - T +31 (0)315 555 465
Order : 112048.145	Remarks : PLC controlled panel	Revision by : KRM	Check : PAV	
Engineer : KRM	Title : Front page	Revision date : 06.04.12	Page : 1 / 78	

Page-index

INDEX_UK

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3	Page index		13.03.13	PAV
4	Spare Page		31.01.11	KRM
5	Spare Page		31.01.11	KRM
6	General instructions		13.03.13	PAV
7	Materials control panel		13.03.13	PAV
8	Wiring colours		31.01.11	KRM
9	Terminal strip identification		10.01.13	RKH
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11	Main wiring		13.03.13	PAV
11.1	Main wiring		10.01.13	RKH
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21	Main wiring		10.01.13	PAV
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27	PLC configuration		20.03.12	KRM
28	PLC overview		28.12.12	BAD
29	PLC overview		28.12.12	BAD
30	PLC overview		28.12.12	KRM
31	PLC overview		28.12.12	KRM
32	PLC overview		28.12.12	KRM
33	PLC overview		28.12.12	KRM
34	PLC overview		28.12.12	KRM
35	PLC overview		13.03.13	KRM
36	PLC wiring		10.01.13	RKH
37	PLC wiring		10.01.13	RKH
38	PLC wiring		10.01.13	RKH
39	PLC wiring		10.01.13	RKH
40	PLC wiring		28.12.12	PAV
41	PLC wiring		28.12.12	PAV
42	PLC wiring		28.12.12	PAV
43	PLC wiring		28.12.12	PAV
44	PLC wiring		28.12.12	PAV
45	PLC wiring		28.12.12	PAV
46	PLC wiring		28.12.12	PAV
47	PLC wiring		13.03.13	PAV
48	PLC wiring		06.04.12	KRM
49	Control wiring		10.01.13	PAV



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Project : Austral Ilo - 112048	Revision : A	Created on : 17.07.06
Order : 112048.145	Revision by : PAV	Check : PAV
Engineer : KRM	Revision date : 13.03.13	Page : 2 / 78
Remarks : PLC controlled panel		
Title : Page index		



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Project : Austral Ilo - 112048		Revision : A	Created on : 17.07.06
Order : 112048.145	Remarks : PLC controlled panel	Revision by : PAV	Check : PAV
Engineer : KRM	Title : Page index	Revision date : 13.03.13	Page : 3 / 78

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Spare page

Project : Austral Ilo - 112048			Revision : A	Created on : 17.07.06	 VISION ON SUSTAINABILITY Copyright reserved www.nirxuis-water.com - T +31 (0)315 955 466				
Order : 112048.145	Remarks : PLC controlled panel		Revision by : KRM	Check : PAV					
Engineer : KRM	Title : Spare Page		Revision date : 31.01.11	Page : 5 / 78					

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Cabling

- * Cable diameters can not be fully determined by NWT since cable type, cable length, installation method, local conditions and local regulations are unknown parameters.
- * The given cable diameters are based on the following:
 - Cable insulation: PVC
 - Wire material: Copper
 - Cable length: Max. 100 meter
 - Ambient temperature: 30°C
 - No direct sunlight exposure
 - NON hazardous area
- * Cable diameters stated in the cable connection diagrams are therefore only valid for cables according above criteria.
At the end NWT can not be held responsible for the installed cables, the end responsibility is for the local contractor!
- * Be aware: cables must be selected in conformance with power consumption, environmental conditions, method of installation and local regulations.
- * Mentioned cable brands and types are NWT standard. The customer is free to choose a preferred equivalent of another brand or type.

Local Isolators

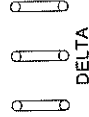
- * For safety, NWT strongly advises to install local isolators for each motor.
 - Local isolators are not in NWT scope of supply.
 - Local isolators have to be supplied by local cable contractor.

Motor connections Star or Delta (Be Aware, Y or D connection VSD controlled motors can be different!)

- * Motor plate is indicating 230/400V - D/Y
motor must be connected in STAR

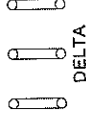


Motor plate is indicating 400/690V - D/Y
motor must be connected in DELTA



Motor connections Star or Delta for frequency controlled motors

- * Small frequency controlled motors (<0.55kW), are powered up with 3 x 230VAC and must therefore be connected in Delta!



Project : Austral Ilo - 112048

Order : 112048.145 Remarks : PLC controlled panel

Engineer : KRM Title : General instructions

Revision : A Created on : 17.07.06

Revision by : PAV Check : PAV

Revision date : 13.03.13 Page : 6 / 78

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
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Materials Control Panel

Component	Brand	Type/remark
Control Panel	Eldon / Rittal	Coated steel Dimensions (WxHxD) :
Fan	Eldon / Rittal	EF series
Main switch	ABB	OT 250A-E4
Emergency relay	Telemecanique	XPS-AF5130
Motor protections	Allen Bradley	140M-C2E series
Motor relays	Allen Bradley	100-C series
Miniature Circuit Breakers	Allen Bradley	1492-series
Switches	Telemecanique	Switches KAC1H + K1B001U
Lamps	Telemecanique	ZB5 (LED type)
Interface Relais	Finder	46.61...
Power Supply 24VDC	Siemens	PS307
Terminals	Phoenix	PIT/PIT-MT
Fused terminals	Phoenix	PIT/PIT-MT
PLC	Siemens	S7-313C series
Display	Exor UniOP	Elop 307

Project : Austral Ilo - 112048		Revision : A	Created on : 17.07.06
Order : 112048.145	Remarks : PLC controlled panel	Revision by : PAV	Check : PAV
Engineer : KRM	Title : Materials control panel	Revision date : 13.03.13	Page : 7 / 78



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Colours Wiring Control Panel :

Purpose of Wiring	Wire Type	Colour
400VAC Main power circuit	L1	Black
	L2	Black
	L3	Black
	Neutral	Blue
	Earth	Green/Yellow striped
230VAC Main power circuit	Phase	Black
	Neutral	Blue
230VAC UPS supplied circuit	Phase	Orange (1.5mm ²)
	Neutral	Orange (1.5mm ²)
230VAC Control voltage circuit	230VAC-A	Black/red
	230VAC-B	Blue/red
24VAC Control voltage circuit	24VAC-A	Red
	24VAC-B	Red/white
24VDC Control voltage circuit	Phase	Blue
	Neutral	Blue/white
(Analog) measuring signals	4-20mA signals, PTC & PT100 protections	White
Potential free wiring power supplied by others	Signal and Return	Orange (1mm ²)

Project : Austral Ilo - 112048		Revision : A	Created on : 17.07.06
Order : 112048.145	Remarks : PLC controlled panel	Revision by : KRM	Check : PAV
Engineer : KRM	Title : Wiring colours	Revision date : 31.01.11	Page : 8 / 78



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Terminal strip identification

Strip	Title
X1	X1= Main power supply
X2	X2= Motor groups
X3	X3= Emergency stop
X4	X4= Analog input signals
X5	X5= Analog output signals
X6	X6= Temperature protections (PTC)
X7	
X8	X8= Temperature protections (PT100)
X9	
X10	X10= Process connections group 1
X11	X11= Process connections group 2
X12	X12= Process connections group 3
X13	
X14	
X15	
X16	
X17	
X18	
X19	
X20	X20= Contacts central alarm
X100	
X101	X101= External start/stop signals
X102	X102= Local isolators

Project : Austral Ilo - 112048

Order : 112048.145

Engineer : KRM

Remarks : PLC controlled panel

Title : Terminal strip identification

Revision : A

Revision by : RKH

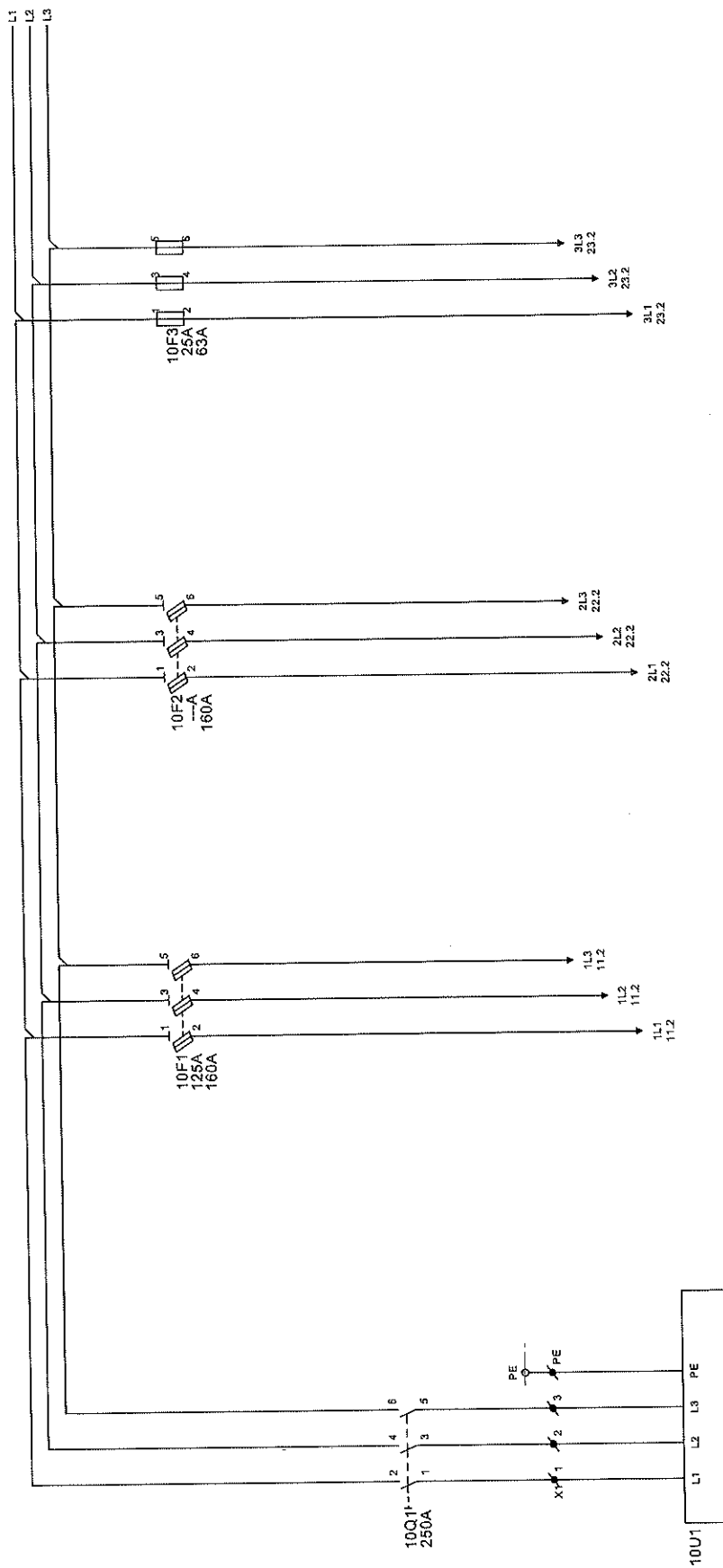
Revision date : 10.01.13

Created on : 17.07.06

Check : PAV

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Power Supply
3x440V 60Hz
Total installed power : Approx. 50kW
Advised pre-fuse : 200A

Power supply
motor groups
3 phase 400VAC
Total load approx. 108A

Power supply
motor groups
3 phase 400VAC
Spare installed
for future

Power supply
control voltage

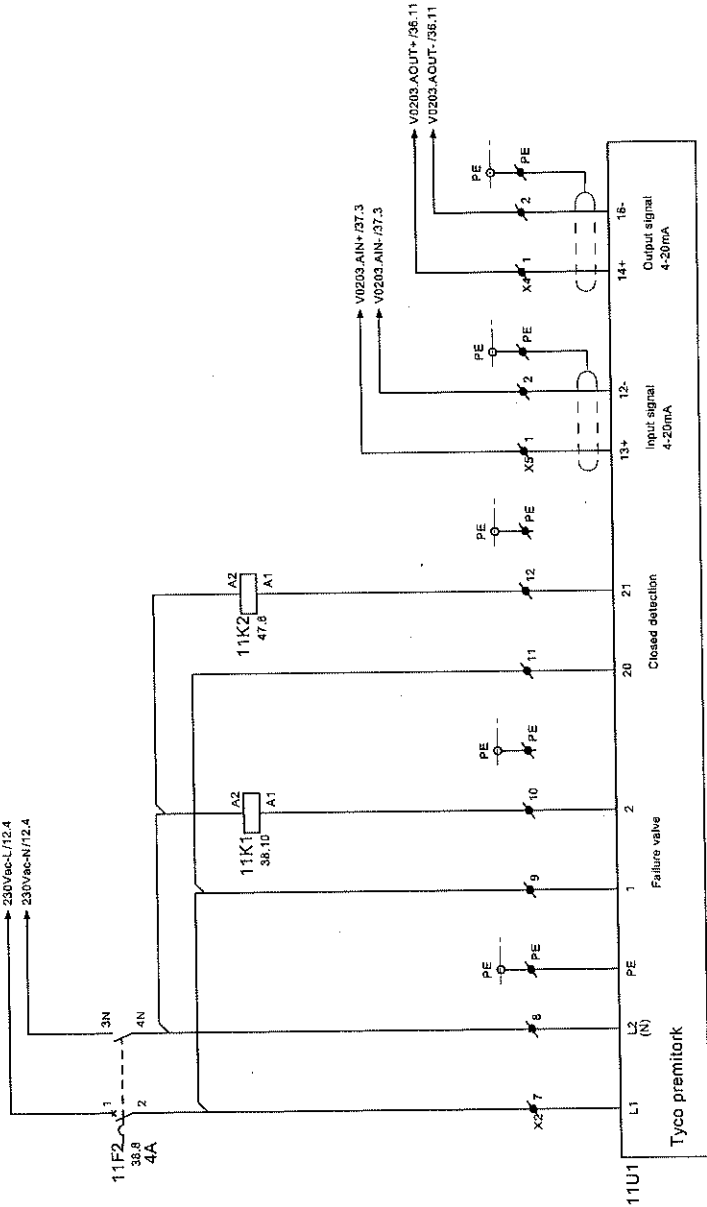


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Revision	: A	Created on	: 17.07.06
Revision by	: PAV	Check	: PAV
Revision date	: 13.03.13	Page	: 10 / 78

Project	: Austral Ilo - 112048
Order	: 112048.145
Remarks	: PLC controlled panel
Engineer	: KRM
Title	: Main wiring

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Influent control valve
flotation unit
V0203

Detection failure
control valve
V0203

Detection
closed
V0203

Position control.
control valve
V0203

Position meas.
valve
V0203



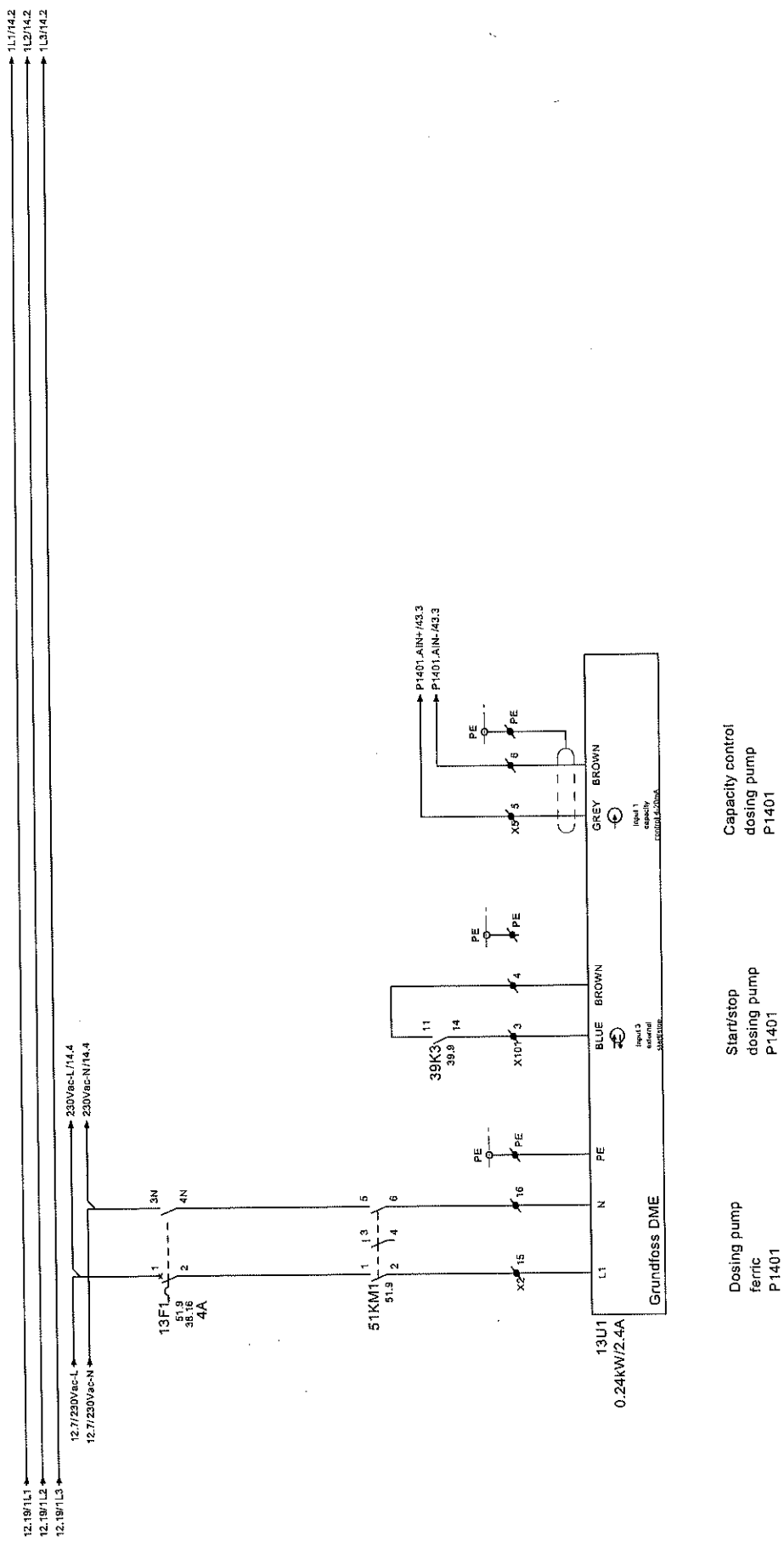
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Project : Austral Ilo - 112048	Revision : A	Created on : 17.07.06
Order : 112048.145	Revision by : RKH	Check : PAV
Engineer : RKH	Revision date : 10.01.13	Page : 11.1 / 78

Remarks : PLC controlled panel

Title : Main wiring

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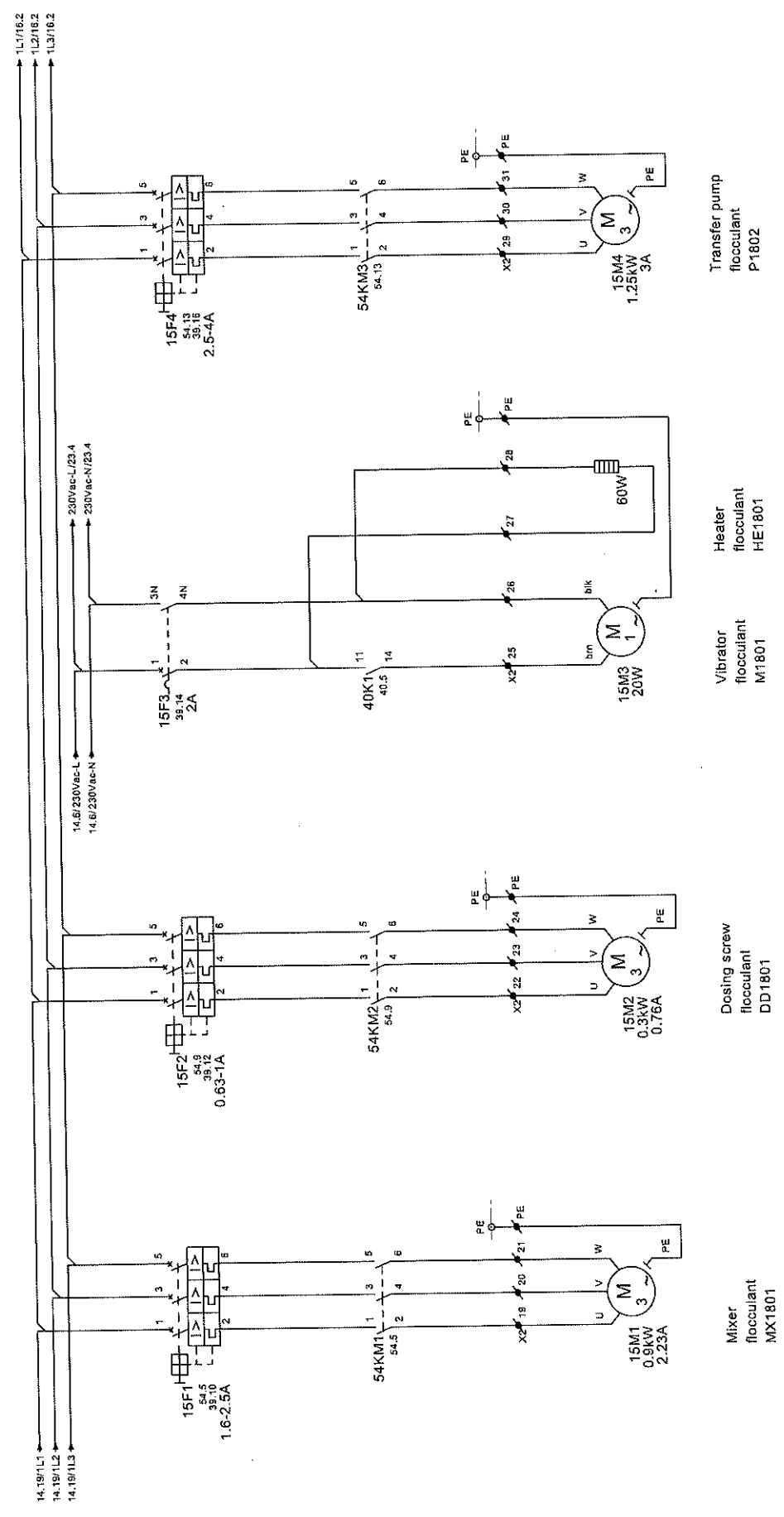
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Order : 112048.145	Remarks : PLC controlled panel	Revision by : PAV		Check : PAV	
Engineer : KRM	Title : Main wiring	Revision date : 10.01.13		Page : 13 / 78	

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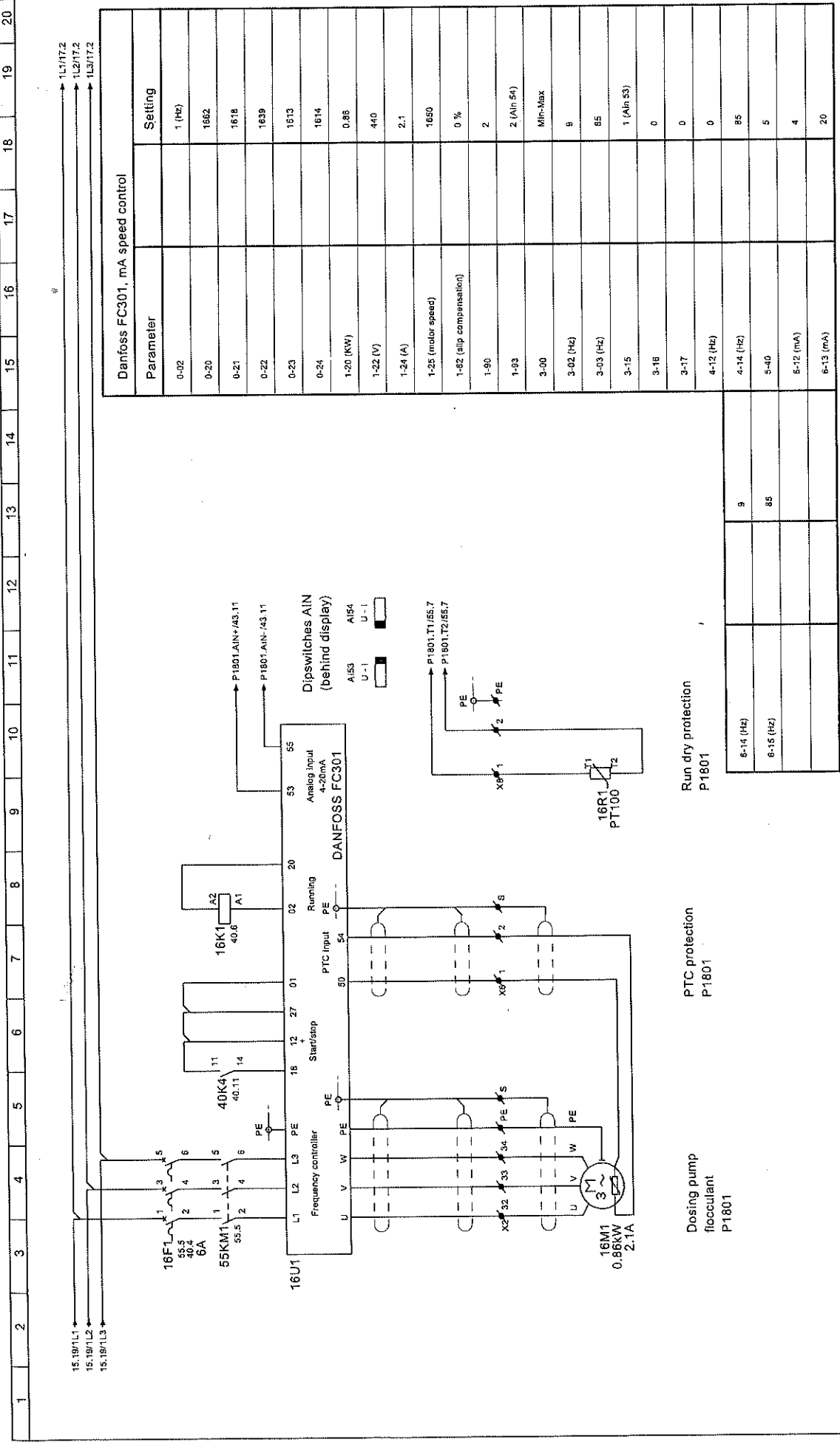
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Project : Austral Ilo - 112048	Revision : A	Created on : 17.07.06
Order : 112048.145	Revision by : PAV	Check : PAV
Engineer : KRM	Revision date : 10.01.13	Page : 15 / 78
Remarks : PLC controlled panel		Title : Main wiring



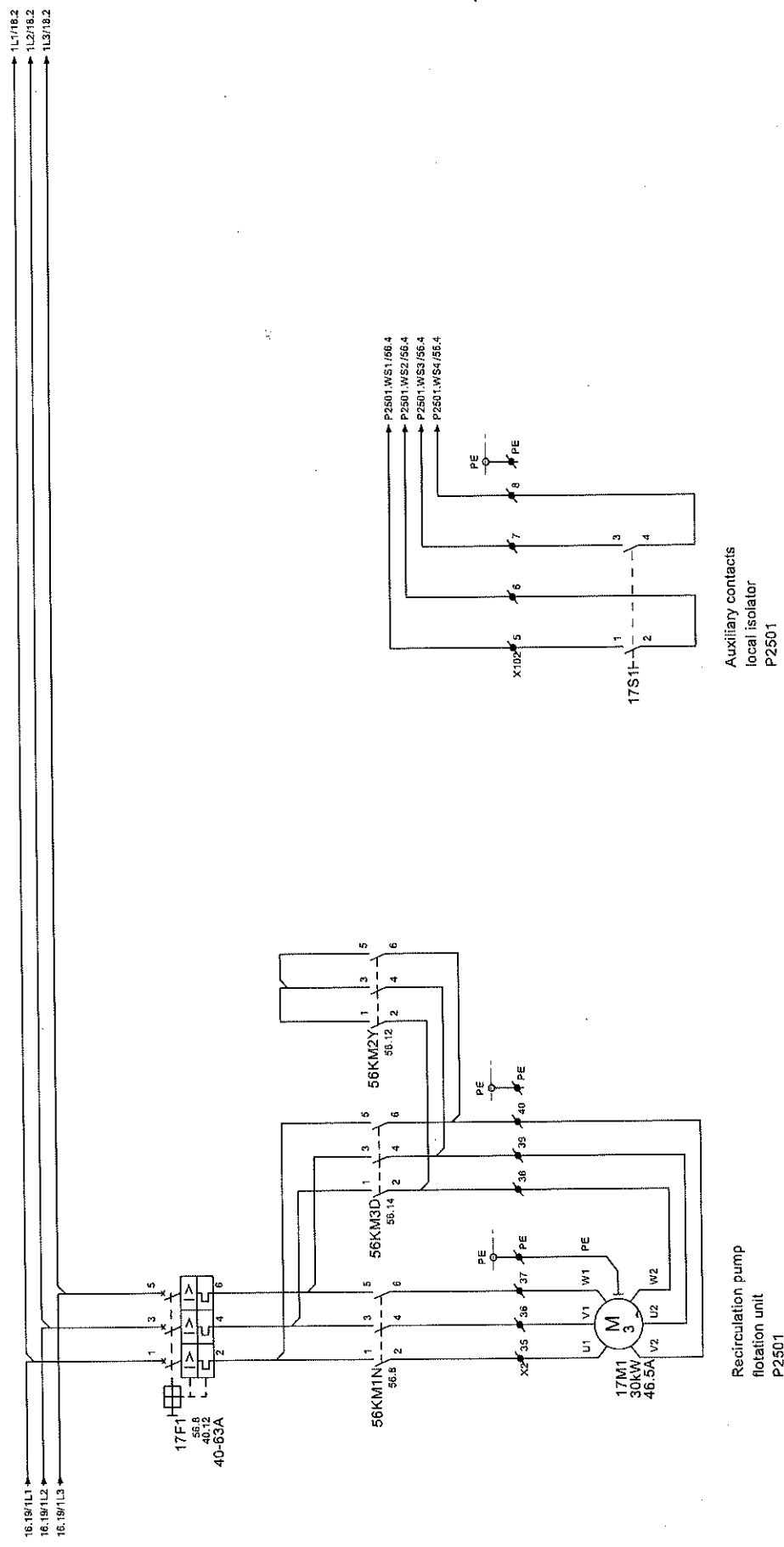
Danfoss FC301, mA speed control		
Parameter		Setting
0-02		1 (Hz)
0-20		1662
0-21		1618
0-22		1639
0-23		1613
0-24		1614
1-20 (kW)		0.86
1-22 (V)		440
1-24 (A)		2.1
1-25 (molar speed)		1650
1-52 (slip compensation)		0 %
1-90		2
1-93		2 (Ain 54)
3-00		Min-Max
3-02 (Hz)		9
3-03 (Hz)		85
3-15		1 (Ain 53)
3-16		0
3-17		0
4-12 (Hz)		0
4-14 (Hz)		85
5-40		5
6-12 (mA)		4
6-13 (mA)		20

Project : Austral Ilo - 112048		Revision : A	Created on : 17.07.06
Order : 112048.145		Revision by : PAV	Check : PAV
Engineer : KRM		Revision date : 10.01.13	Page : 16 / 78
Remarks : PLC controlled panel		Title : Main wiring	



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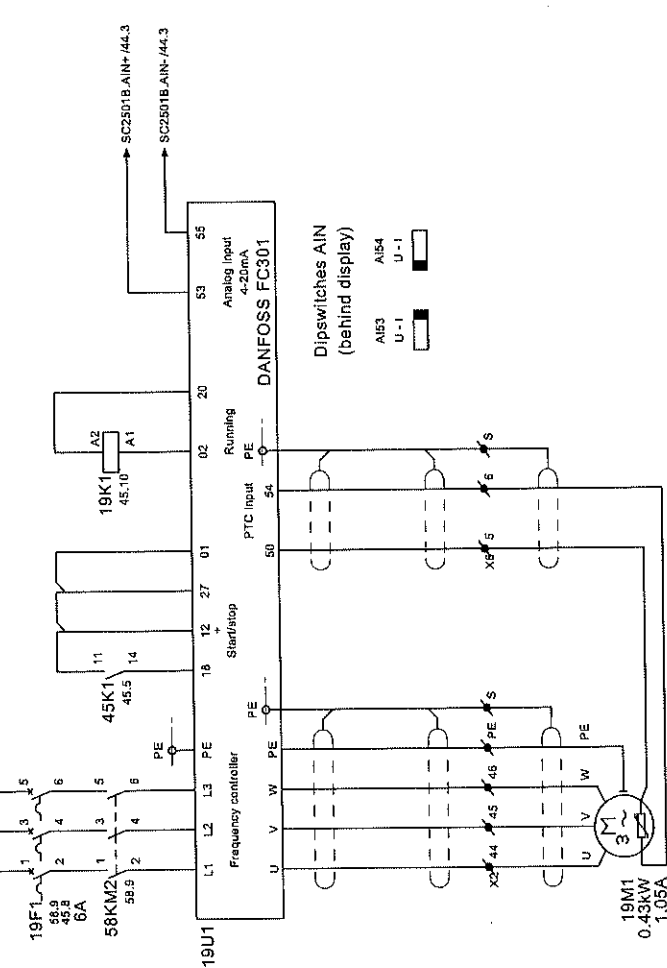


Project : Austral Ilo - 112048		Revision : A	Created on : 17.07.06
Order : 112048.145		Revision by : RKH	Check : PAV
Engineer : KRM		Revision date : 10.01.13	Page : 17 / 78
Remarks : PLC controlled panel		Title : Main wiring	

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Danfoss FC301, mA speed control		
Parameter		Setting
0-02		1 (Hz)
0-20		1602
0-21		1618
0-22		1599
0-23		1613
0-24		1614
1-20 (kW)		0.43
1-22 (V)		440
1-24 (A)		1.05
1-90		2
1-93		2 (Ain 54)
3-00		Min-Max
3-02 (Hz)		20
3-03 (Hz)		50
3-15		1 (Ain 53)
3-16		0
3-17		0
4-12 (Hz)		0
4-14 (Hz)		50
5-40		5
6-12 (mA)		4
6-13 (mA)		20
6-14 (Hz)		20
6-16 (Hz)		50

Top skimmer 2
floatation unit
SC2501B

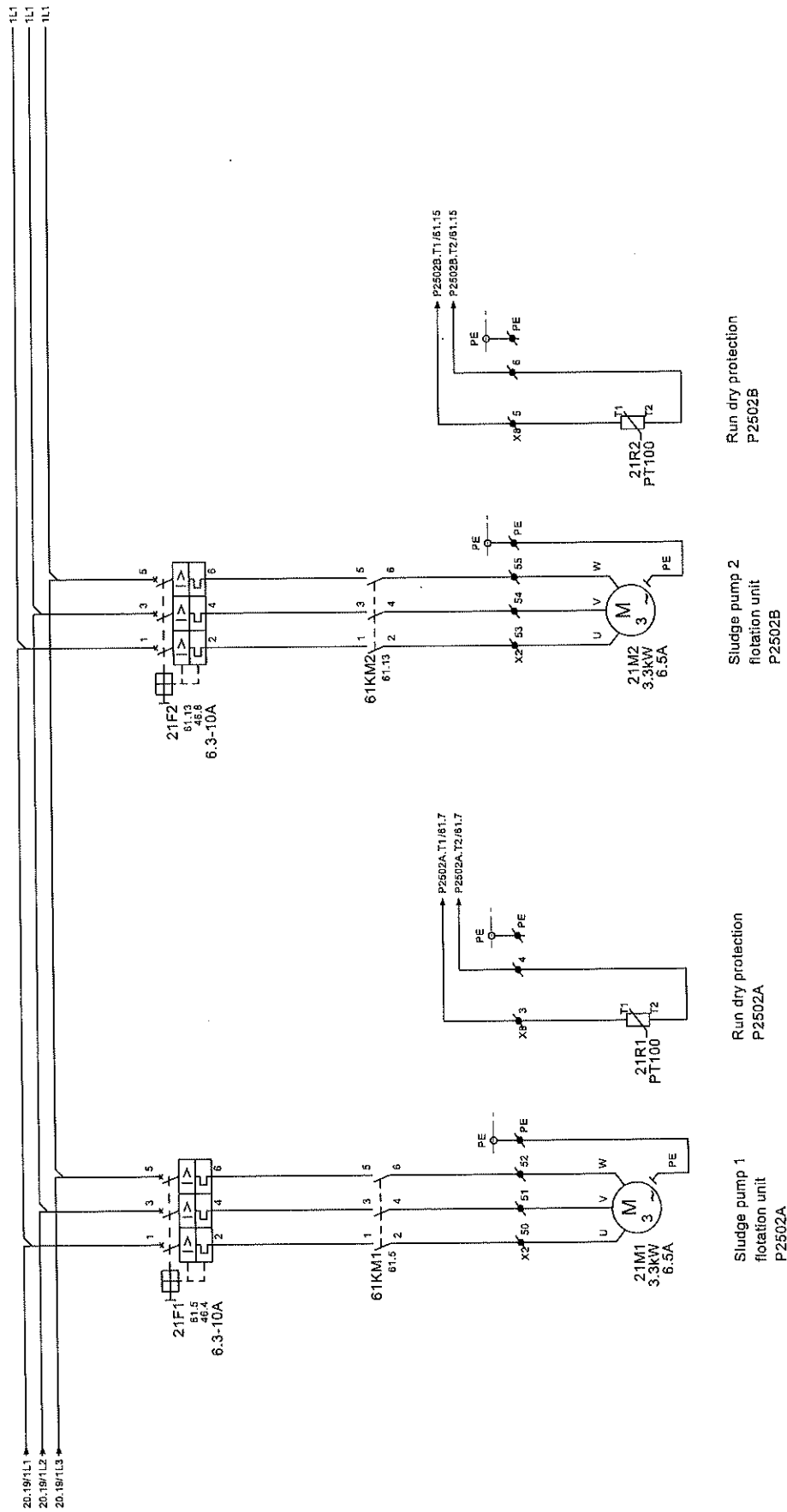
PTC protection
SC2501B



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Project : Austral Ilo - 112048	Revision : A	Created on : 17.07.06
Order : 112048.145	Revision by : PAV	Check : PAV
Engineer : KRM	Revision date : 10.01.13	Page : 19 / 78
Remarks : PLC controlled panel		
Title : Main wiring		

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Project : Austral Ilo - 112048		Revision : A		Created on : 17.07.06	
Order : 112048.145		Remarks : PLC controlled panel		Check : PAV	
Engineer : KRM		Title : Main wiring		Page : 21 / 78	



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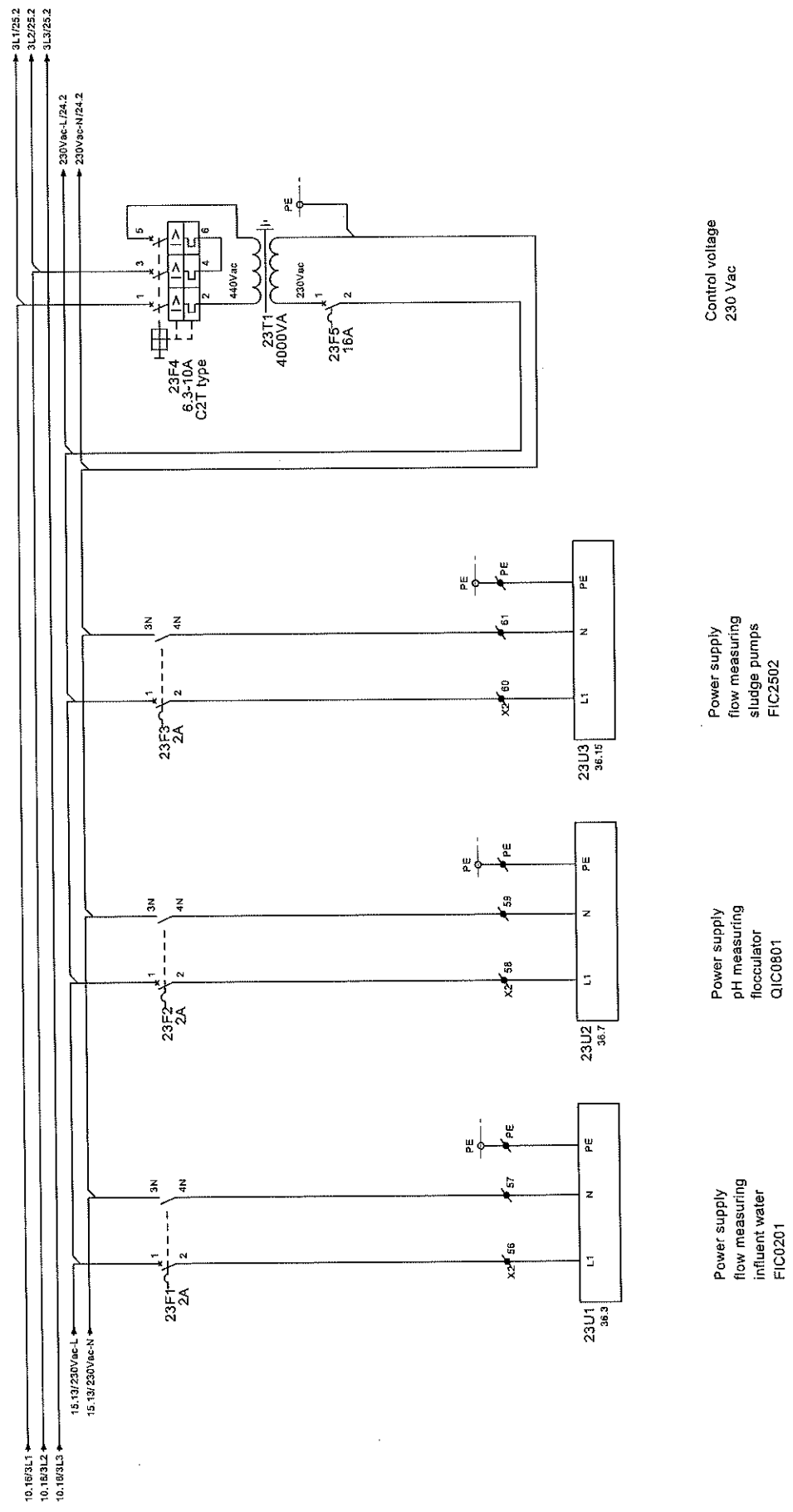
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10.12/21.1	21.1
10.12/21.2	21.1
10.12/21.3	21.1

Spare page

Project : Austral Ilo - 112048		Revision : A	Created on : 17.07.06
Order : 112048.145	Remarks : PLC controlled panel	Revision by : BAD	Check : PAV
Engineer : KRM	Title : Main wiring	Revision date : 28.02.12	Page : 22 / 78
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Power supply
flow measuring
inflow water
FIC0201

Power supply
pH measuring
floculator
QIC0801

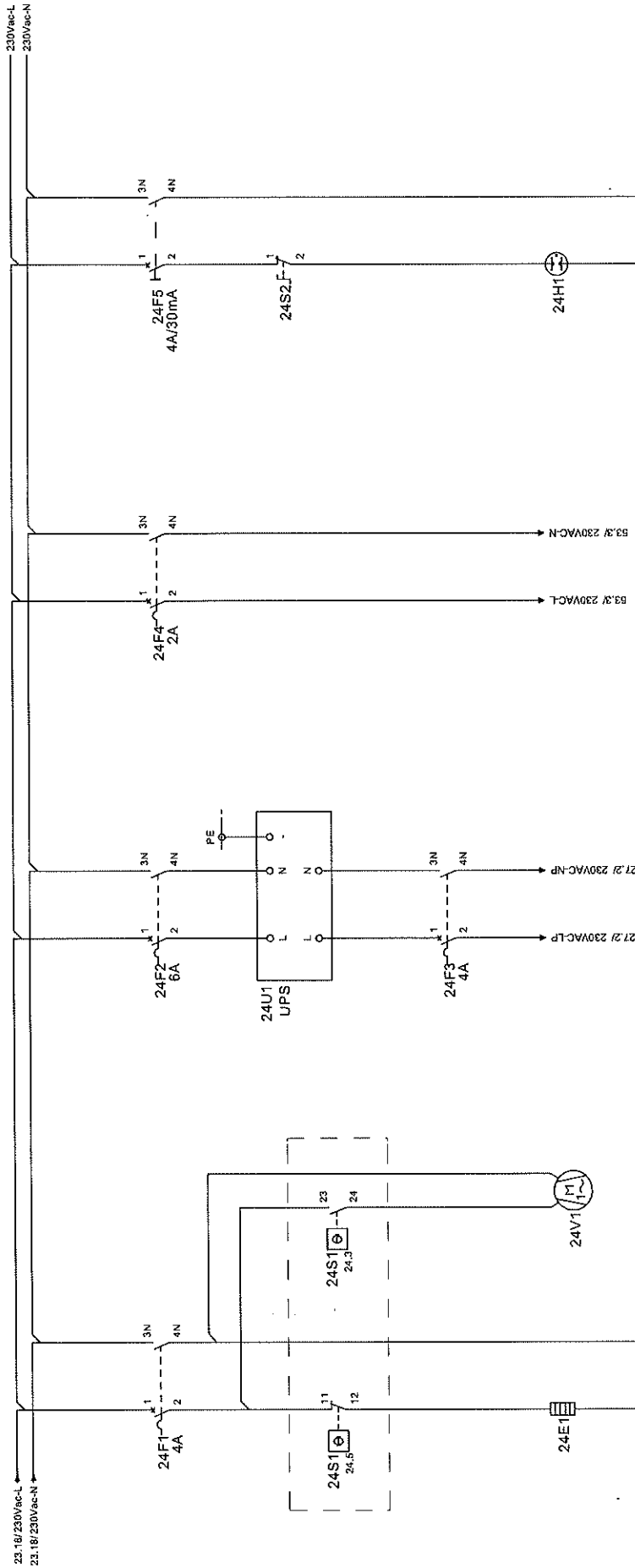
Power supply
flow measuring
sludge pumps
FIC2502



Project : Austral Ilo - 112048	Revision : A	Created on : 17.07.06
Order : 112048.145	Revision by : PAV	Check : PAV
Engineer : KRM	Revision date : 10.01.13	Page : 23 / 78

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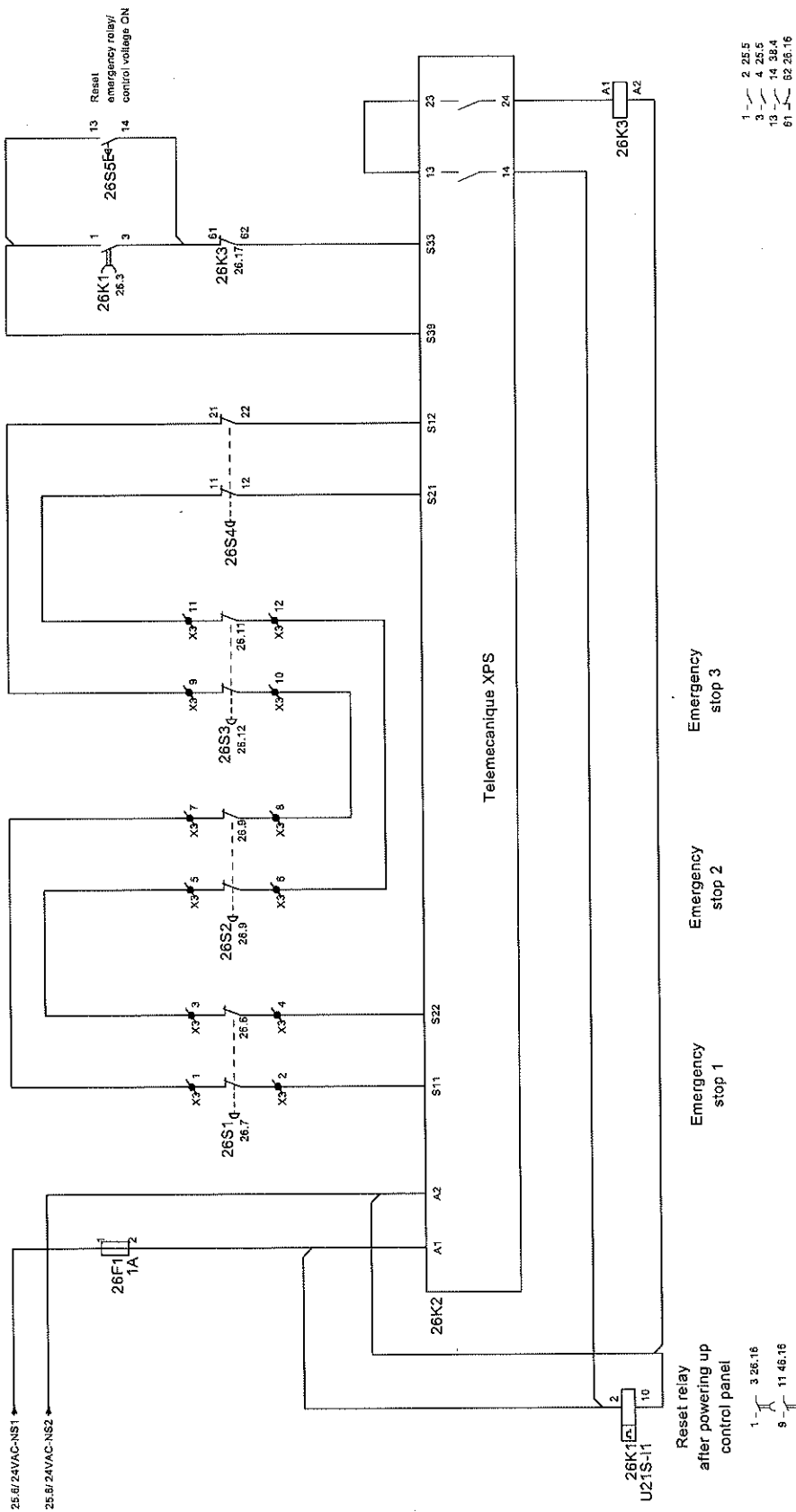
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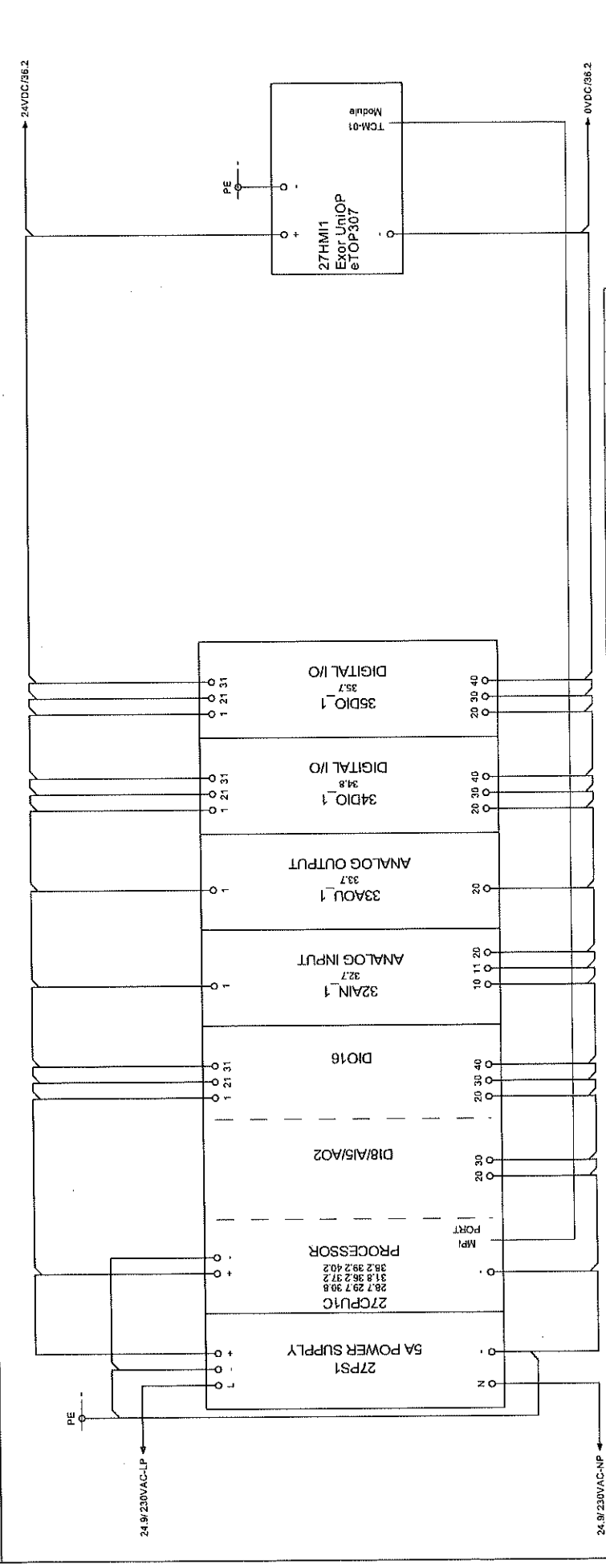
Heating control panel Fan control panel Power supply PLC & HMI Power supply PT100, Carlo Cavazzi & PTC relays Panel lighting with door switch & wall socket 230VAC for programming device

Project : Austral Ilo - 112048		Revision : A	Created on : 17.07.06	 VISION ON SUSTAINABILITY Copyright reserved www.nihiluis-water.com - T +31 (0)315 855 466
Order : 112048.145	Remarks : PLC controlled panel	Revision by : KRM	Check : PAV	
Engineer: KRM	Title : Main wiring	Revision date : 28.12.12	Page : 24 / 78	

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
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Project : Austral Ilo - 112048		Revision : A	Created on : 17.07.06
Order : 112048.145		Revision by : KRM	Check : PAV
Engineer : KRM		Revision date : 28.12.12	Page : 26 / 78
Remarks : PLC controlled panel		Title : Emergency stop	



6ES7 390-1A30-0AA0 MOUNTING RAIL	6ES7 392-1AM00-0AA0 POWER SUPPLY	6ES7 313-6EB01-0AB0 CPU 313C	6ES7 953-8LF- MMC 128KB	6ES7 392-1AM00-0AA0 CONNECTOR 40P (2x)	6ES7 331-7KF- 8X ANALOG INPUT	6ES7 392-1AJ- CONNECTOR 20P	6ES7 332-5HF- 8X ANALOG OUTPUT	6ES7 392-1AM- CONNECTOR 40P	6ES7 323-1BL00-0AB0 16X DIGITAL I/O	6ES7 392-1AM00-0AA0 CONNECTOR 40P	6ES7 323-1BL00-0AB0 16X DIGITAL I/O	6ES7 392-1AM00-0AA0 CONNECTOR 40P	6ES7 392-1AM00-0AA0 CONNECTOR 40P	6ES7 901-0BF00-0AA0 MPI CABLE	6ES7 972-0BB-... PROFIBUS CONNECTOR
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27CPU1C
27.4
AI5

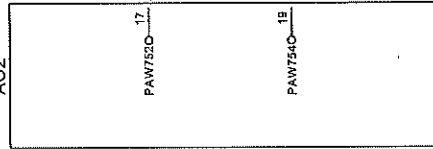
PEW7520	3	35.4	Flow measuring influent water flotation unit FIC0201
PEW7540	6	35.8	pH measuring flocculator C1C0801
PEW7560	8	35.12	Position measuring influent control valve flotation unit V0203
PEW7580	12	35.16	Flow measuring sludge pumps flotation unit FIC2502

Analog Input
Siemens S7-300
AI5/AO2/DI24/DO16
6ES7 313-5BE01-0AB0
CPU313C

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27CPU1C
27.4
AO2



Analog output
Siemens S7-300
AI5/AO2/DI24/DO16
6ES7 313-6BE01-0AB0
CPU313C

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27CPU1C
27.4
DIO16

Start level preparing flocculant LS1801 39.4	2	22	A124.0C	39.5	Feed pump floatation unit P0201
High level preparing flocculant LS1801 39.6	3	23	A124.1C	39.7	Dosing pump coagulant P1201
Flow counter flocculant station FS1801 39.8	4	24	A124.2C	39.9	Dosing pump ferric P1401
Motor protection mixer flocculant MX1801 39.10	5	25	A124.3C	39.11	Safety valve coagulant V1411
Motor protection dosing screw flocculant DO1801 39.12	6	26	A124.4C	39.13	Dosing pump neutraliser P1501
Circuit breaker vibrator flocculant M1801 39.14	7	27	A124.5C	39.15	Safety valve neutraliser V1511
Motor protection transfer pump flocculant P1802 39.16	8	28	A124.6C	39.17	Filling water valve flocculant make up unit V1803
Det. low level flocc. dosing compartment LS1802 39.18	9	29	A124.7C	39.19	Mixer flocculant MX1801
Circuit breaker dosing pump flocculant P1801 40.4	12	32	A125.0C	40.5	Vibrator flocculant M1801
Running detection dosing pump flocculant P1801 40.5	13	33	A125.1C	40.7	Dosing screw flocculant DO1801
Run dry protection dosing pump flocculant P1801 40.8	14	34	A125.2C	40.9	Transfer pump flocculant P1802
Flow switch post dilution flocculant FS1802 40.10	15	35	A125.3C	40.11	Dosing pump flocculant P1801
Motor protection recirculation pump floatation unit P2501 40.12	16	36	A125.4C	40.13	Solenoid valve post dilution V1813
Auxiliary contacts local isolator P2501 40.14	17	37	A125.5C	40.15	Recirculation pump floatation unit P2501
Pressure switch recirculation pump PS2501 40.16	18	38	A125.6C	40.17	Aeration valve floatation unit V2513
Pressure switch compressed air PS2502 40.18	19	39	A125.7C	40.19	Top skimmer 1 floatation unit SC2501A

Digital input/output
Siemens S7-300
AI5/AO2/DI24/DO16
6ES7 313-5BE01-0AB0
CPU313C



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Title : PLC overview		

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27CPU1C
27.4
DI8

Emergency relay activated 38.4 —O E126.0

Motor protection feed pump P0201 38.6 —O E126.1

Circuit breaker influent control valve flotation unit V0203 38.8 —O E126.2

Detection failure control valve V0203 38.10 —O E126.3

Circuit breaker dosing pump P1201 38.12 —O E126.4

Detection low level coagulant LS1201 38.14 —O E126.5

Circuit breaker dosing pump ferric P1401 38.16 —O E126.6

Circuit breaker dosing pump neutraliser P1501 38.18 —O E126.7

Digital Input
Siemens S7-300
AI5/AO2/DI24/DO16
6ES7 313-5BE01-0AB0
CPU313C

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33AOU_1
27.3
44.2

PAW2720	3	43.4	Desired capacity dosing pump P1401
PAW2740	7	43.8	Desired capacity dosing pump P1501
PAW2760	11	43.12	Frequency controller dosing pump P1801
PAW2780	15	43.16	Frequency controller top skimmer 1 SC2501A
PAW2800	23	44.4	Frequency controller top skimmer 2 SC2601B
PAW2820	27	44.8	Frequency controller bottom skimmer SC2602
PAW2840	31	44.12	Spare analog output
PAW2860	35	44.16	Spare analog output

Analog output
Siemens S7-300
8 x 4-20mA
6ES 7332-5HF-0AB0
Slot 5

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34DIO_1
27.10
45.2
46.2

Circuit breaker top skimmer 1 floatation unit SC2501A 45.4	2	22	A8.00	45.5	Top skimmer 2 floatation unit SC2501B
Run detection freq. controller SC2501A 45.6	3	23	A8.10	45.7	Bottom skimmer floatation unit SC2502
Circuit breaker top skimmer 2 floatation unit SC2501B 45.8	4	24	A8.20	45.9	Sand drain 1 floatation unit V2505A
Run detection freq. controller SC2501B 45.10	5	25	A8.30	45.11	Sand drain 2 floatation unit V2505B
Circuit breaker bottom skimmer floatation unit SC2502 45.12	6	26	A8.40	45.13	Sand drain 3 floatation unit V2505C
Run detection freq. controller SC2502 45.14	7	27	A8.50	45.15	Sand drain 4 floatation unit V2505D
Low level detection ferlic LS1401 45.16	8	28	A8.60	45.17	Sludge drain 1 floatation unit V2505A
Low level detection neutralisant LS1501 45.18	9	29	A8.70	45.19	Sludge drain 2 floatation unit V2505B
Motor protection sludge pump 1 floatation unit P2502A 45.4	32	32	A9.00	45.5	Sludge drain 3 floatation unit V2505C
Run dry protection P2502A 45.6	33	33	A9.10	45.7	Stop feeding balance tank due to high level LC0201 (to client)
Motor protection sludge pump 2 floatation unit P2502B 45.8	34	34	A9.20	45.9	Spare
Run dry protection P2502B 45.10	35	35	A9.30	45.11	Spare
Flow pulses flow measuring influent water floatation unit FIC0201 45.12	36	36	A9.40	45.13	Manual operation active
Detection control voltage 24Vac 45.14	37	37	A9.50	45.15	Contact Central Alarm
Reset Central Alarm 45.16	38	38	A9.60	45.17	Lamp Red Alarm Signal
Reset Buzzer 45.18	39	39	A9.70	45.19	Buzzer Alarm Signal

Digital In-/Output
Siemens S7-300
16 Input & 16 Output
6ES7 323-1BL00-0AA0
Slot 6

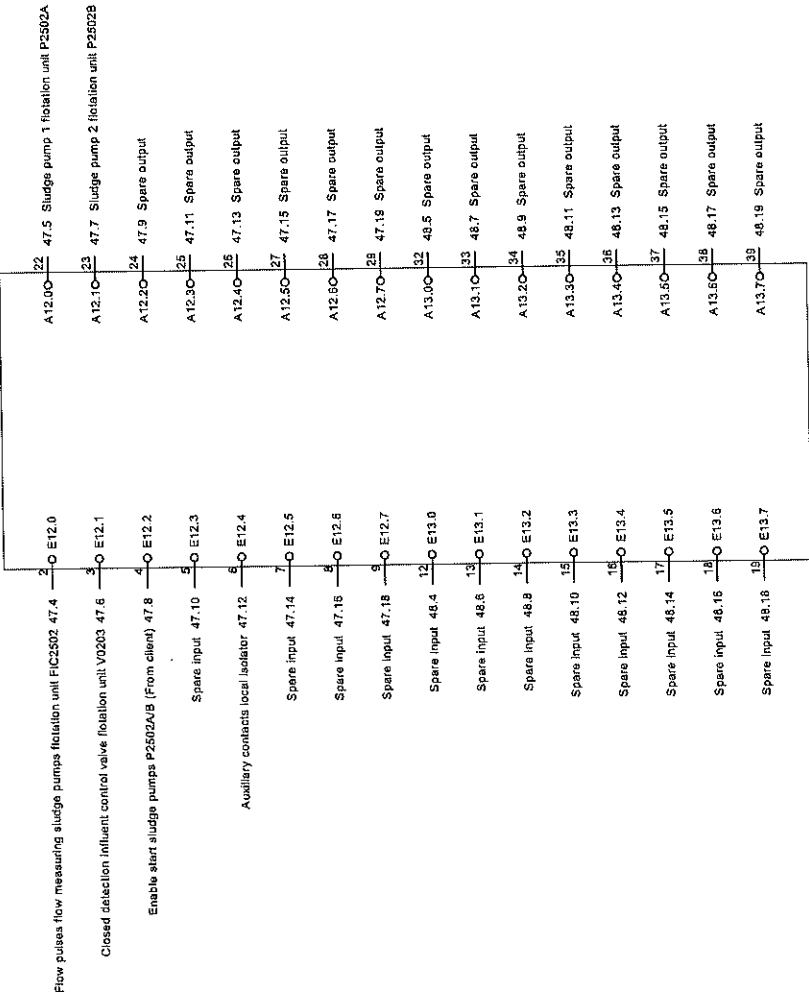
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35DIO_1
27.11
47.2
48.2



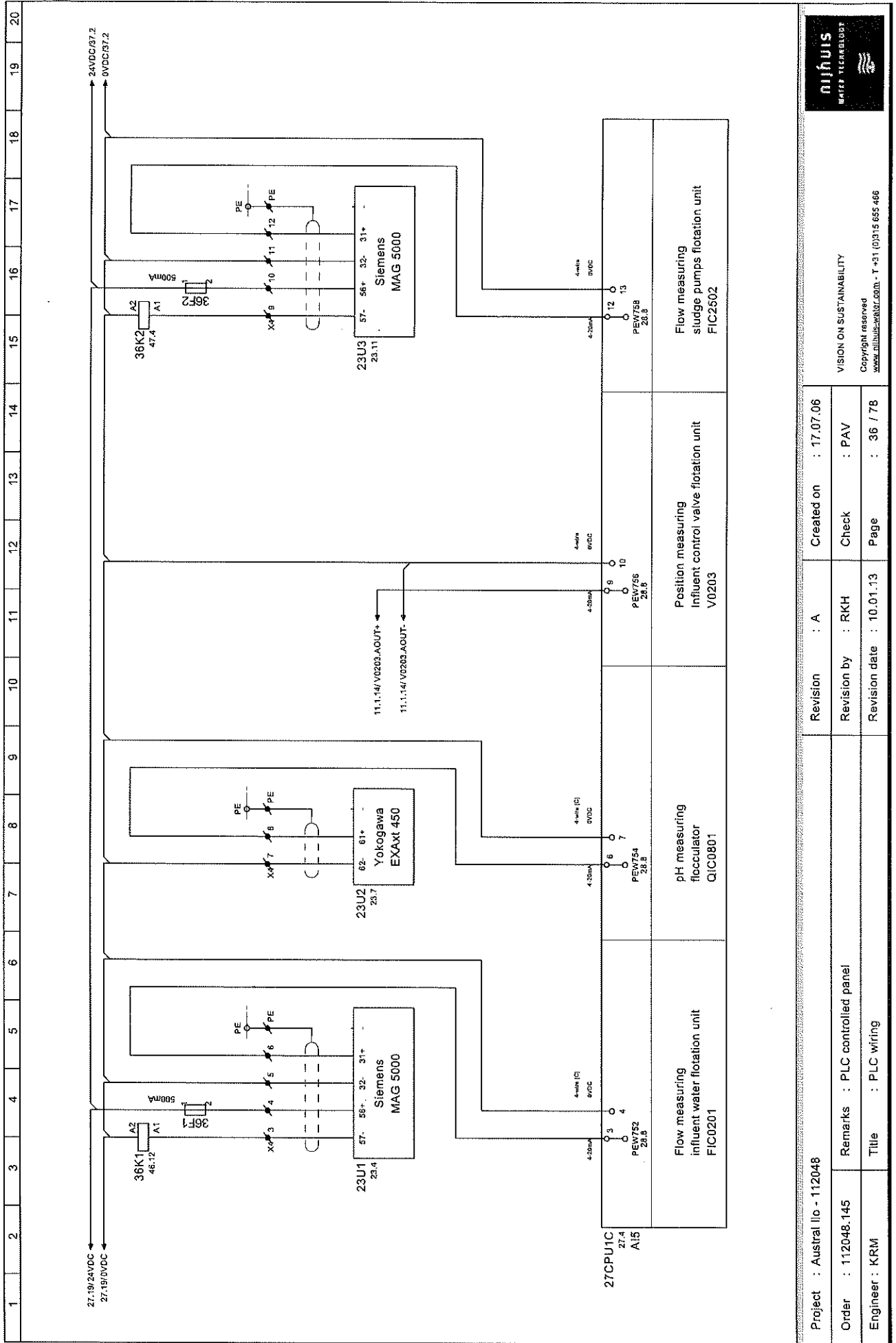
Digital In-/Output
Siemens S7-300
16 Input & 16 Output
6ES7 323-1BL00-0AA0
Slot 7



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Remarks : PLC controlled panel
Title : PLC overview



Project : Austral Ilo - 112048

Order : 112048.145

Engineer : KRM

Remarks : PLC controlled panel

Title : PLC wiring

Revision : A

Revision by : RKH

Revision date : 10.01.13

Created on : 17.07.06

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12.11/P1201.AIN-

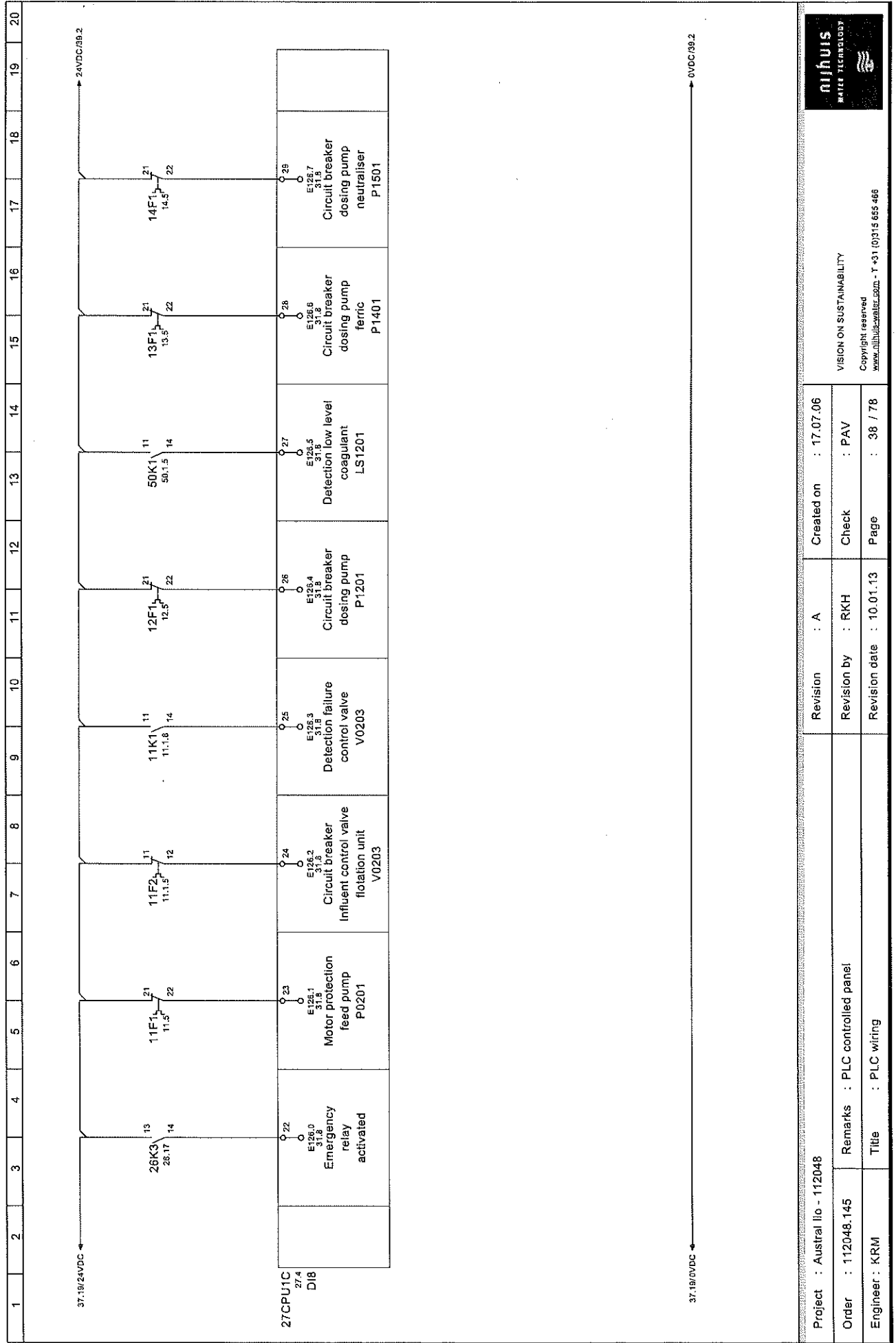
12.11/P1201.AIN+

27CPU1C 27.4 AO2	Q 17 PAW752 28.8	Q 19 PAW754 28.8
Desired setpoint Influent control valve flotation unit V0203	Desired capacity dosing pump P1201	

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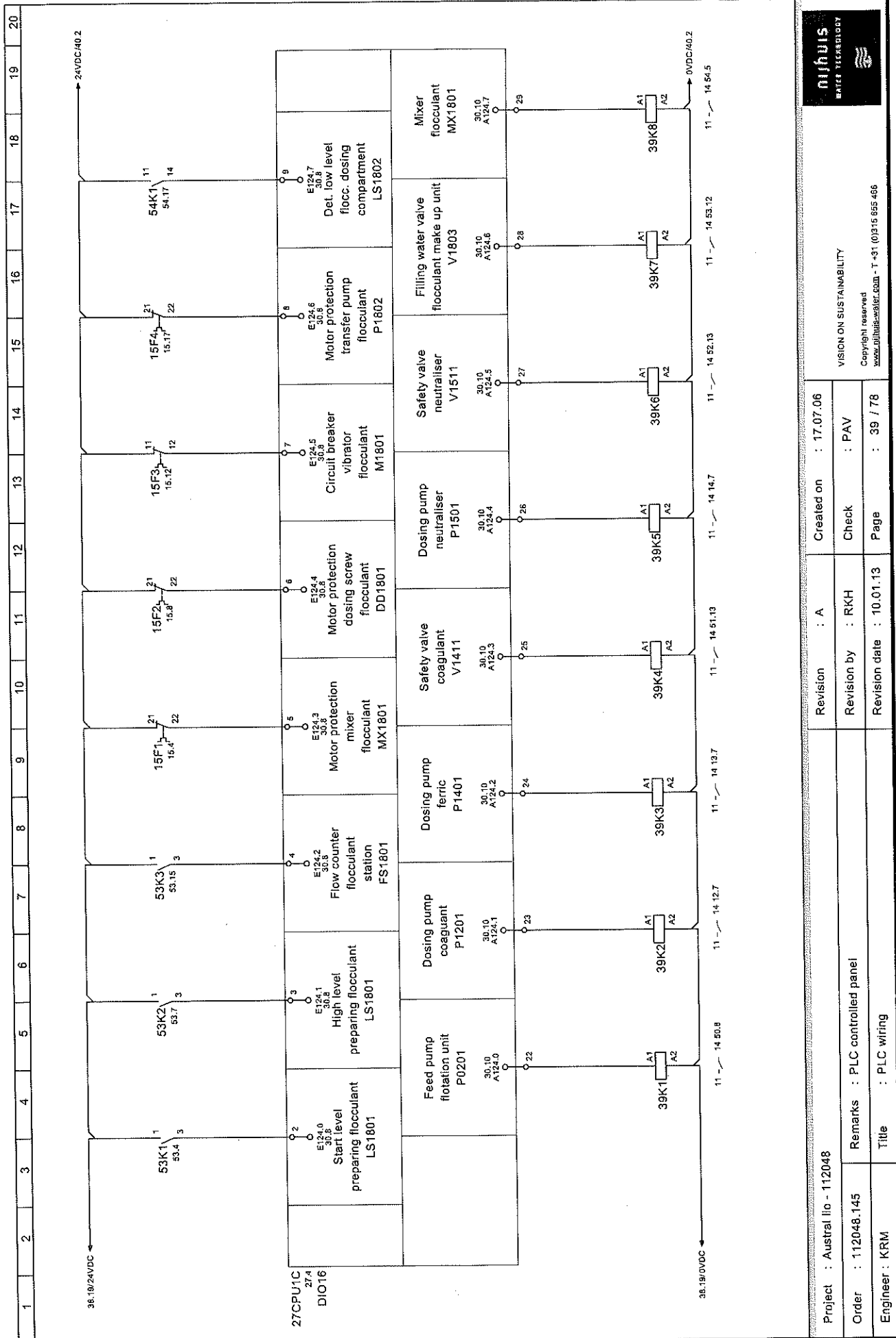


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